

Coarse-to-fine Estimation: Compressive Sensing for High-Resolution Inverse SAR Imaging

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Abstract

Compressive sensing (CS) enables imaging with less data, assuming sparsity in certain domains. We proposed a ground-breaking coarse-to-fine (CTF) estimation method that reconstructs images with fewer signal supports over a fine grid. Owing to the explicit storage and usage within sparse reconstruction, the sensing matrix (SM) is resource-intensive, even for small problem sizes. To alleviate this issue, the CTF procedure discards a certain number of columns from the SM as the reconstruction progresses from coarser to finer scales. A comprehensive experimental validation was performed using ISAR datasets of both real targets and targets of known geometry, and the qualitative and quantitative metrics of the imaging results confirmed the efficacy of retrieving high-resolution ISAR images. We substituted a single SM with an exorbitant size with a sequence of SMs of reduced size, thereby reducing computational demands and making the technique suitable for real-time applications. The CTF method reconstructed the target's image that was free from clutter and effectively decoupled the required number of measurements from the ISAR data, thus achieving super-resolution. The entropy of the reconstructed image in the fine stage proves the completeness of the imaging.

1 Introduction

Imaging from sparse data plays a pivotal role in radar target detection technology [1]. Methods such as range-Doppler imaging, polar formatting, matched filtering, and sparse reconstruction are used to reconstruct ISAR images [2]. By leveraging Compressive Sensing (CS) theory, it is possible to obtain high-quality images using notably less data when compared with conventional imaging approaches that require full data. CS-based ISAR imaging has gained attention owing to its ability to achieve high-quality results with a reduced number of samples in radar applications [3]. This objective can be realized by exploiting sparsity assumptions pertaining to the scene or target within specific domains. CS techniques have been employed in radar image processing, including ISAR imaging, SAR imaging of dynamic targets, spotlight SAR imaging of static objects, and SAR imaging of oceanic targets [4].

Recent studies have focused on exploiting 1D sparsity for ISAR imaging with rotational motion compensation. The ISAR phase compensation and estimation of the target motion parameters were integrated into the CS imaging framework. However, this leads to the dislocation of adjacent echo pulses in the range dimension owing to range compression [5, 6]. In addition, azimuthal phase compensation is required to align the echoes for turntable imaging. Furthermore, the complex nonlinear motion of the target renders image reconstruction challenging.

Our primary goal is to enhance the quality of ISAR image reconstruction within the CS framework. This enhancement can be achieved by utilizing a novel coarse-to-fine

estimation approach to alleviate the computational burden. Various techniques, such as basis pursuit [7], matching pursuit [8], and thresholding-based methods [9], can be employed to achieve CS reconstruction through the resolution of l_1 -minimization problems. Orthogonal Matching Pursuit (OMP), a greedy algorithm, has gained substantial traction in the domain of SAR and ISAR imaging owing to its efficiency and robustness [10].

Our method focuses on enhancing the target imaging using 2D SM with a CTF approach that entails a set of coarseness factors ($X(1/F) = 2^{2F}$). The reconstruction process involves progressively restricting the signal support, thus eliminating a specific number of columns from the SM as the algorithm transitions from one factor to another. The proposed technique does not require a pre-processing chain and introduces a direct reconstruction solution that strategically considers both the computational cost, memory gain, and imaging resolution. The CTF strategy discards the unwanted frequencies in the fine stage, reconstructs the image with fewer samples, and achieves a high resolution with negligible clutter.

However, to accurately assess the capability of CS to reconstruct high-resolution ISAR images, it is imperative to conduct a comparative analysis against established conventional techniques [11]. To fulfill this requirement, comparisons were made between images reconstructed using traditional wave numbers ($\omega - k$) and the proposed ISAR imaging method [12]. This comparison was subsequently quantified using numerical metrics. The reconstructed ISAR images were evaluated using indicators such as Entropy and Mean Absolute Error (MAE) or Δ_{error} and further

assessed through quantitative measures such as the Target-to-Clutter Ratio (TCR), Root Mean Square Error (RMSE), and Size Overlap Ratio [13, 14].

Section 2 presents ISAR imaging models and proposed coarse-to-fine estimation. Section 3 presents both qualitative and quantitative performance indicators. Section 4 presents the imaging results of the proposed coarse-to-fine estimation and conventional ISAR methods. Finally, Section 5 presents the conclusions of this research and future perspectives.

2 Signal Processing for ISAR

This section elucidates the CS signal model, introduces a CTF estimation, and delineates a robust greedy algorithm tailored for the enhancement of CS-ISAR imaging.

2.1 Traditional CS Imaging

CS theory asserts that when a vector represents a signal of interest, it can be precisely reconstructed using a limited set of linear, non-adaptive measurements, given that the vector displays sparsity on an orthonormal basis or in a certain dictionary. The ISAR image comprises the locations and amplitudes of the strong echoes. Let the sparse echo contain P pulses and the fully observed signal contain M pulses, with each pulse containing N sampling points, and the under-sampled model of the sparse aperture is provided by $P \ll M$. The classic CS measurement model in the form of a linear system is given as follows [15]:

$$\mathbf{Y} = \mathbf{A}\mathbf{S}, \text{ where, } \mathbf{A} = \Phi\Psi \text{ so, } \mathbf{Y} = \Phi\Psi\mathbf{S} \quad (1)$$

where $\mathbf{S}$ represents the sparse data of the ISAR measurements (vectorized) to be reconstructed, and ' $\mathbf{Y}$ ' is the vectorized version of the measurements. Ψ and Φ are Fourier transforms in range and azimuth dimensions, respectively. The sensing matrix ' $\mathbf{A}$ ' is a 2D square matrix as defined below in eq. (2). The measurements were free of motion compensation or range compression. The ISAR problem is considered an ill-posed inverse problem. SM corresponds to the image reconstruction model. For 2D image reconstruction, the sensing model implies two Fourier transforms in $\mathbf{A}$ that correspond to the ISAR imaging model, as follows:

$$\mathbf{A} = \mathcal{F}_N \otimes \mathcal{F}_N \quad (2)$$

where $\otimes$ denotes Kronecker product, and $\mathbf{A} \in \mathbb{C}^{N^2 \times N^2}$, and $\mathcal{F}_N$ is a $N \times N$ discrete Fourier transform (DFT) matrix [16]. However, for reconstruction, the $\mathbf{A}$ matrix is subjected to a different coarseness factor of 2^{2F} . A CTF scheme exploits the spatial sparsity of $\mathbf{S}$ to screen out regions with dense information [17]. This reduces memory requirements and increases the robustness of the reconstruction.

2.2 Proposed CTF Estimation Technique

SM plays a pivotal role in CS imaging; however, its explicit storage and usage within sparse reconstruction procedures are resource-intensive, even for small problem sizes. As a solution, we propose a groundbreaking CTF estimation method for the reconstruction of ISAR images with restricted cardinalities. Let $X(F)$ be the mask representation, where $F \in \{1, 2, 4\}$ presents the different coarseness

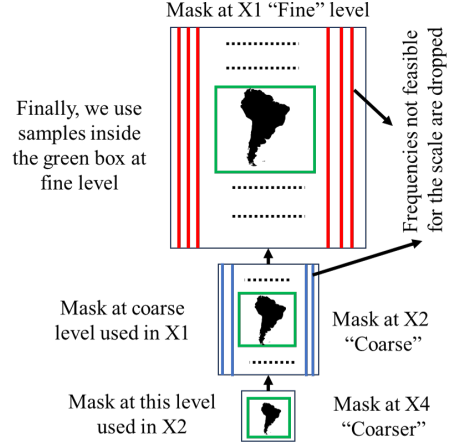


Figure 1 CTF estimation, propagation of the target mask from the coarser to coarse and then fine stages allows effectively pruning the feasible signal support at X1.

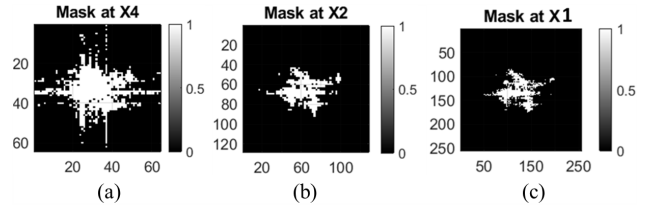


Figure 2 Masks obtained for Yak-42 by CTF estimation at different scales: (a) mask obtained at coarser level X4, (b) mask obtained at coarse level X2, and (c) final mask obtained at fine level X1.

factors (**Coarser, Coarse, Fine**).

Let $\Omega^{(i)} = \{j_i, X_j \neq 0\}$ denote the support set (active column indices) at the i^{th} scale. The next support set Ω^{i+1} scale from the i^{th} scale is given as follows:

$$\Omega^{(i+1)} = \cup_{j \in \Omega^{(i)}} \{2j; 2j-1; 2j+N; 2j+N-1\} \quad (3)$$

For vectorizing the 2D space into 1D, it goes first over rows and then over columns, where $n_{\text{rows}} \times n_{\text{columns}}$ denotes the domain size, while $n_{\text{rows}} = n_{\text{columns}} = N$.

Fig. 1 shows the scheme for CTF estimation restricted to the feasible signal-support regions. The mask at the coarser level (X4) was obtained by identifying the signal support, which was then transformed into an initial mask for the next coarse level (X2), and only the feasible signal supports were preserved at a fine level (X1). Thus, the fine stage utilizes a few samples from the sparse vector $\mathbf{S}$ and forces the clutter regions to be zero. The image reconstructed at the fine stage attained a resolution higher than the exploited bandwidth as there were frequency gaps.

Fig. 2 shows the process for selecting the mask for the Yak-42 target. Initially, a mask is generated with the coarsest resolution image using a specific threshold ' th ', as shown in Fig. 2(a). This mask was then adapted to a coarse level to effectively reduce image support over a large grid, as shown in Fig. 2(b). Subsequently, this coarse-level mask was introduced during the fine stage of the process, as shown in Fig. 2(c). Here, we observed a further reduction in the allowable image support compared to the larger grid used in the fine stage. This approach starts with a coarser level to minimize the memory requirements of

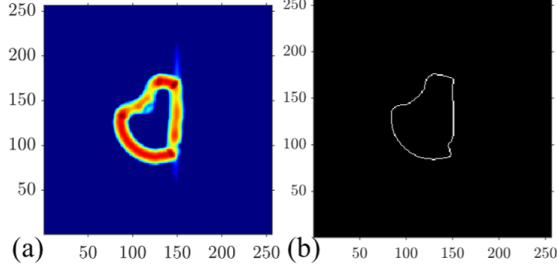


Figure 3 Image processing to obtain the actual size of the target: (a) averaged ISAR image ρ and (b) final edge or boundary of the target extracted by thresholding.

the SM while reaching a finer scale, which consequently works for sparse measurements and reconstructs images.

2.3 Orthogonal Matching Pursuit (OMP)

The OMP algorithm updates the sparse signal ($\mathbf{S}$) approximation in each iteration (T) by projecting $\mathbf{Y}$ onto the columns of the sparsifying matrix using the current support set while minimizing the l_2 norm of the residuals ($\mathbf{R}$) and avoiding re-selection of entries, as follows:

$$\mathbf{S}^{(T)} = \mathbf{A}_{\Omega^{(T)}}^\dagger \mathbf{Y} \text{ and } \mathbf{R} = \mathbf{Y} - \mathbf{A}_{\Omega^{(T)}} \mathbf{S}^{(T)} \quad (4)$$

where $\mathbf{R}$ is the residual vector, the difference between the input and the reconstructed signal, while $\dagger$ denotes Moore-Penrose transpose. The hit-and-trial technique, involving dictionary changes and iteration adjustments, typically enhances sparsity reconstruction [18].

3 Imaging Performance Indicators

We compared images reconstructed using the traditional $\omega - k$ and CTF CS-based ISAR methods both qualitatively and quantitatively.

3.1 Qualitative Indicator

The target image contours from both the conventional method and the CTF X1 stage were overlapped for size comparison. The reconstructed image I is convoluted and averaged over an $N \times N$ box car filter, as follows:

$$\rho = \langle |I| \otimes |I| \rangle \quad (5)$$

where $\langle \cdot \rangle$ is the ensemble averaging. The average image shown in Fig. 3(a) is subjected to a threshold ‘ th ’ to obtain a binary logical image, as follows:

$$I_L(i, j) = 1 \iff \rho(i, j) > th \quad (6)$$

A logical image I_L fills the gaps in the grayscale image to extract the outer boundaries and yields aligned perimeters that match the target size, as shown in Fig. 3(b).

3.2 Quantitative Indicator

Two distinct datasets are exploited, comprising three ground truth targets and the Yak-42 dataset, to demonstrate the efficacy of our CTF image reconstruction technique. The ground truth targets encompass cylindrical, rectangular, and non-canonical shapes, with the fourth being Yak-42 aircraft data. To quantitatively evaluate the reconstructed image quality, we calculated certain metrics: target-to-clutter ratio (TCR), root mean square error

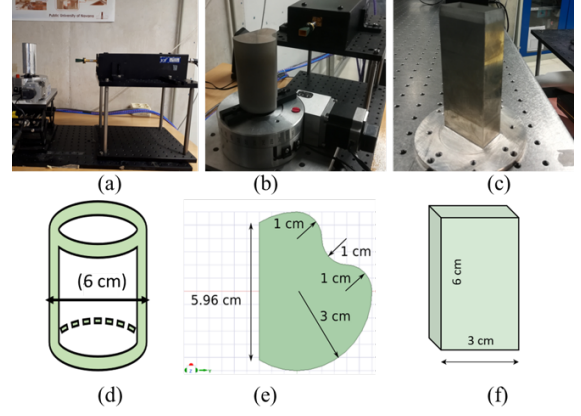


Figure 4 Ground truth (GT) targets: (a–c) cylindrical, Non-canonical, and Rectangular targets with (d–f) geometrical models, respectively.

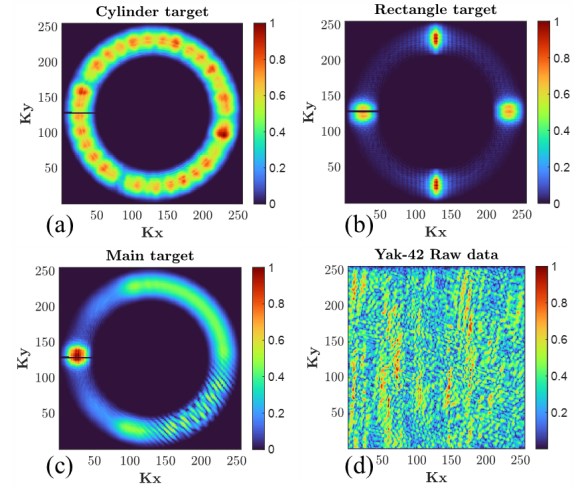


Figure 5 In (a), (b), and (c), the raw data of GT targets exists in the k -space of a dimension of 256×256 . (d) in the case of the Yak-42, we take a step back from its range-compressed data to obtain entirely raw data.

(RMSE), and image entropy [19]. The TCR is defined as:

$$\text{TCR} = 10 \cdot \log_{10} \left(\frac{\sum_{(i,j)} |I_T(i, j)|^2}{\sum_{(i,j)} |I_C(i, j)|^2} \right) \quad (7)$$

where (i, j) is the pixel index of the reconstructed image I and the subscripts T and C correspond to the target and clutter pixels, respectively. The entropy value is a parameter used to demonstrate the image quality and is given as:

$$\text{Entropy} = - \sum_{(i,j)} \frac{|I(i, j)|^2}{E} \log_2 \frac{|I(i, j)|^2}{E} \quad (8)$$

where E denotes the total energy of the reconstructed image I . The entropy image E_I at (i, j) is estimated as:

$$E_I(i, j) = - \sum_k \left(\frac{p_k}{\sum_l p_l} \right) \cdot \log_2 \left(\frac{p_k}{\sum_l p_l} \right) \quad (9)$$

where p_k is the probability distribution of pixel values in the local neighborhood around pixel (i, j) , summation are over k values in the neighborhood, and p_l is sum of all

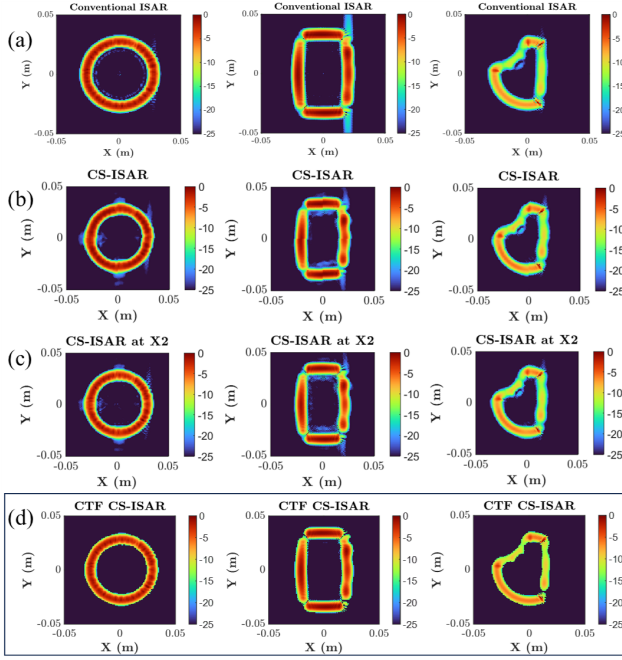


Figure 6 Imaging results for Ground Targets: (a) $\omega - k$ ISAR imaging, (b) CS-imaging using SM of Doppler sequences (at X2), (c) CTF CS-ISAR imaging with LS (at X2), (d) CTF CS-ISAR imaging with LS (X1).

the probabilities. The mean absolute error (MAE) of the reconstructed image I with respect to the GT is given as:

$$\Delta_{\text{error}} = \frac{1}{N} \sum_{(i,j)} |\hat{I}(i,j) - I_{\text{GT}}(i,j)| \quad (10)$$

where $\hat{I}$ and I_{GT} are reconstructed and ground truth images, respectively.

4 ISAR Imaging Results

Fig. 4 shows the ground-truth targets and their geometric models. Fig. 4(a–c) shows cylindrical, non-canonical, and rectangular targets, respectively, with their geometric models in Fig. 4(d–f). The targets were scanned from the middle and positioned in front of the radar system.

The scanning process for our targets operates in stepped frequency mode, where it halts temporarily after a short span of rotation before resuming the rotation. The raw data acquired during this stage necessitates pre-processing for system calibration. This includes tasks such as phase compensation and conversion of data from polar to Cartesian formats. The outcome of this process results in the final raw data being represented in k -space. The measurements were performed on an $N \times N$ -dimensional grid. Fig. 5 (a), (b), and (c) show the k -space data of ground truth targets, while (d) is 2D ISAR raw data of Yak-42.

For GT targets, raw data in the k -space domain enables us to transition from a coarse X2 level to a finer X1 stage. Fig. 6(a) shows the images reconstructed using the traditional $\omega - k$ ISAR imaging method. Fig. 6(b) shows the images reconstructed at the initial X2 level using the SM of the discrete Doppler sequences. Fig. 6(c) shows reconstructions at X2 using the LS method, whereas Fig. 6(d) presents CS imaging at the finer X1 stage reconstructed by

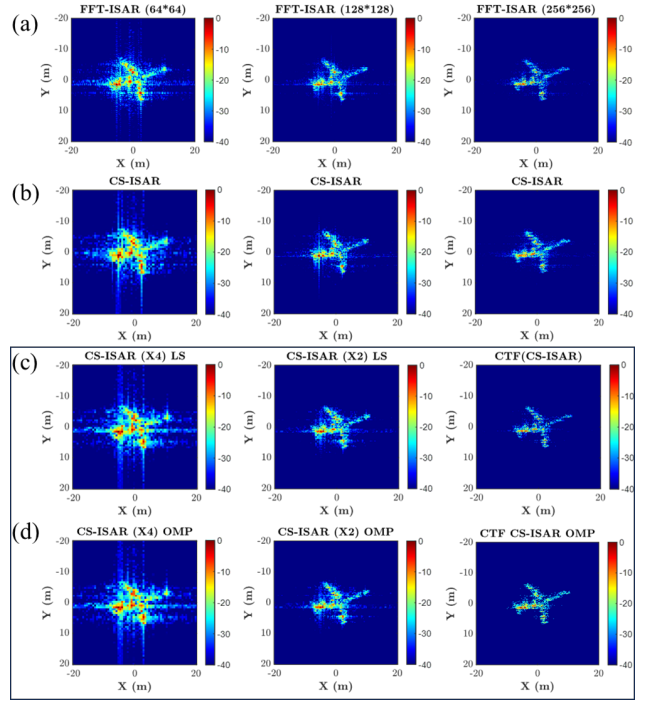


Figure 7 Yak-42 aircraft reconstructed Images: (a) 2D FFT-based imaging, (b) CS-ISAR with Doppler Sequences, (c) CTF CS-ISAR via LS, and (d) CTF CS-ISAR via OMP reconstruction.

reduced measurements. It can be observed that the imaging is clutter-free, as we discarded the unfeasible target support at the fine stage. Table 1 lists the accuracies of the geometric and estimated sizes. These results instill confidence in our CTF estimation for Yak-42 aircraft. We successfully reduced the computational memory demand and enabled high-resolution imaging using the proposed CTF method. Table 2 highlights the ability of the CTF to reconstruct images for finer grids with fewer SM columns, making our approach suitable for real-time applications. Importantly, the challenge of image reconstruction is transformed into a sequential sparse reconstruction.

The imaging results for in-situ data (Yak-42) are presented in Fig. 7, utilizing grids of sizes 64×64 , 128×128 , and 256×256 . Fig. 7(a) shows the reconstructed images obtained using the 2D FFT method without any motion compensation; however, it requires full cardinality. Fig. 7(b)

Table 1 Ground truth Target’s size comparisons obtained by Conventional method and CTF X1 stage.

Target	Original	Conventional	CTF
Cylinder	6	5.8	5.9
Rectangular	6*3	5.9*2.9	5.9*2.9
Non-Canonical	5.96*3	5.94*3	5.95*3

Table 2 Memory gain requirement reduction by eliminating infeasible target support columns (out of 65536).

Cylinder	Rectangular bar	Non Can	Yak-42 LS	Yak-42 OMP
12120	14248	8572	4340	2048

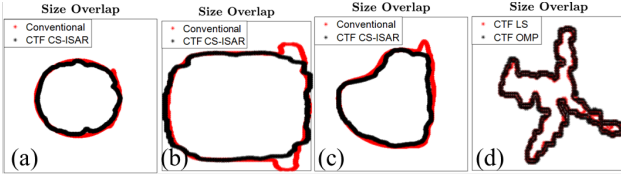


Figure 8 Overlap of target size extracted $\omega - k$ and CTF CS-ISAR images: (a) Cylinder, (b) Rectangular bar, (c) Non-canonical target, and (d) size overlap for Yak-42 X1(LS) & X1(OMP).

Table 3 Overlap ratio of target contour obtained by Conventional and CTF CS-ISAR imaging.

Targets	Overlap ratio of Contours
Cylinder	98.5269
Rectangular	94.2184
Non-Canonical	95.1572
Yak-42	98.6272

shows images reconstructed via CS-ISAR, employing an SM of discrete Doppler sequences, albeit with high computational demands for the full grid. To address this, we applied the proposed CTF technique, leading to image reconstruction using the LS and OMP methods, as shown in Fig. 7(c-d). For the LS approach in the fine stage, as a result of applying the mask, we utilized 4340 (despite 65536) cardinalities from the image vector, whereas the OMP greedy approach employed 2048 cardinalities to achieve high-resolution image reconstruction. We assessed the efficiency of the OMP greedy algorithm in handling image reconstruction at each scale.

In Fig. 8, we conducted a comparative analysis of the overlapping target sizes obtained through conventional ISAR and CTF CS-ISAR imaging. The dimensions obtained using the conventional method are reasonable. However, CTF achieves a clutter-free image, resulting in a clear outline of the target. This validation further underscored the accuracy of the CTF estimation technique. Fig. 8(a-d) display the contour of the target overlap for the cylindrical, rectangular, non-canonical, and Yak-42 targets. For Yak-42, the contours obtained at X1 by LS and X1 by OMP overlapped. We achieved a contour similarity of 95.6% for a complex target. Table 3 provides the overlap ratio of the target contours obtained through both the conventional and CTF methods, confirming the effectiveness of the proposed technique in terms of the geometric size.

Table 4 presents the quantitative performance of the imaging, specifically the TCR and RMSE values. The TCR for conventional ISAR images is reasonable for the given targets. In the case of CTF CS-ISAR, by progressively eliminating unwanted frequency information from X4 to X1, the background becomes clutter-free, resulting in a TCR of Inf, which confirms that the image at the fine stage has a higher resolution owing to zero clutter. The RMSE is promising and sufficient for attaining optimal resolution.

Table 5 presents the Δ_{error} values for both conventional and CTF CS-ISAR images plotted along the X- and Y-axes. These values appear to be reasonable and serve as strong evidence to support the effectiveness of the imaging process. Specifically, the Δ_{error} values for both the

Table 4 Target-to-Clutter-Ratio and Root Mean Square Error for Conventional and CS-ISAR images.

Target/TCR	Conventional	RMSE
Cylinder	12.95	0.0773
Rectangular	12.95	0.2282
Non Canonical	16.41	0.1992
Yak-42	12.63	LS vs OMP 0.0105

Table 5 Δ_{error} between conventional and CTF CS-ISAR images on X and Y axis.

Δ_{error}	Cylinder	Non Canonical	Rect.	Yak-42
$\frac{1}{N} \sum \Delta_X \%$	1.44	1.31	2.24	0.50
$\frac{1}{N} \sum \Delta_Y \%$	1.44	1.31	2.24	0.50

Table 6 Entropy of images obtained by $\omega - k$ and CS-ISAR method.

Target	Conventional	CS-ISAR	CTF
Cylinder	1.72	1.84	1.91
Rectangular	1.25	1.62	2.16
Non-Canonical	1.39	1.45	1.57
Yak-42	1.44	1.68	1.90

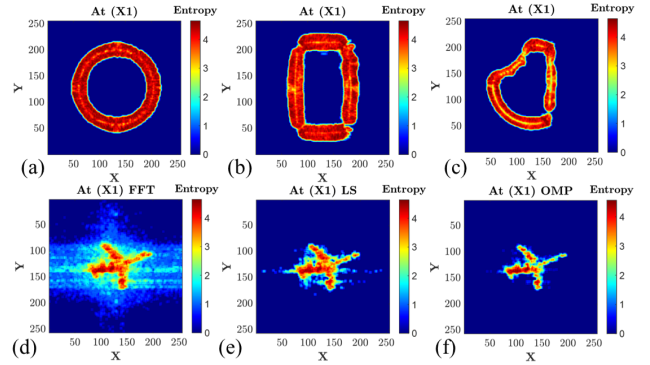


Figure 9 Entropy image at the fine stage. (a), (b), (c) Entropy image of ground-truth targets (d), (e), and (f) Entropy image of Yak-42 reconstructed using FFT, CS-ISAR (LS), and CS-ISAR (OMP), respectively.

ground-truth targets and Yak-42 exhibit validity in accordance with the inherent complexity of the respective targets. The analysis assessed image entropy as a measure of detail, with higher values indicating greater intricacy. Table 6 summarizes the values of image entropy for the conventional CS-ISAR and CTF CS-ISAR methods. The CTF method excelled in preserving the fine details. Fig. 9(a-c) show high entropy in the ground truth targets, confirming accurate fine-stage image reconstruction without clutter. In contrast, Fig. 9(d) reveals the noise in the conventional Yak-42 method. However, Fig. 9(e)-(f) illustrate the significant clutter reduction of the CTF approach, yielding a cleaner Yak-42 representation. Remarkably, using CTF CS-ISAR with OMP and 2048 samples produced a clutter-free image with high entropy, validating its effectiveness. This consistent scale emphasizes the capability of this method.

5 Conclusion

We propose a multiscale CTF estimation method that propagates a mask that restricts the target support across scales, which allows for the restriction of the initial target domain when estimating the ISAR image and the output mask at the next scale. This technique reduces the memory overhead for real-time use in cases where the data samples are limited. Compared to the conventional $\omega - k$ ISAR imaging method, the proposed CTF technique demonstrated its applicability by achieving superior imaging results without the need for any preprocessing. Qualitative and quantitative indicators verified the superior performance of this approach. The target contours obtained by conventional and CTF-CS-ISAR imaging exhibited ambient accuracy. This advancement towards low computational costs and lower memory requirements brings us closer to practical applications.

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